

The fit, and the arithmetic behind it

Chosen model: **Sum of two logistics**, fitted with `scipy.optimize.curve_fit(..., method="lm")`.

$$h(t) = c + a1 / (1 + \exp(-k1(t - t1))) + a2 / (1 + \exp(-k2(t - t2)))$$

Why this model

A single sigmoid cannot come back down. Two logistics have exactly two limbs, with $a2$ negative making the second one a recession.

Fitted parameters

PARAMETER	UNIT	VALUE	SE	T	P-VALUE
c	m	14.3466	0.0129	1115.34	0.00e+00
a1	m	9.8896	1.8568	5.33	2.24e-07
k1	1/h	0.5983	0.0425	14.07	0.00e+00
t1	h	26.3692	0.1610	163.79	0.00e+00
a2	m	-4.6596	1.8631	-2.50	1.30e-02
k2	1/h	0.3744	0.0204	18.33	0.00e+00
t2	h	30.4298	1.3286	22.90	0.00e+00

SSE

2.3531

n = 256, p = 7

SST

1756.0362

m²

R²

0.998660

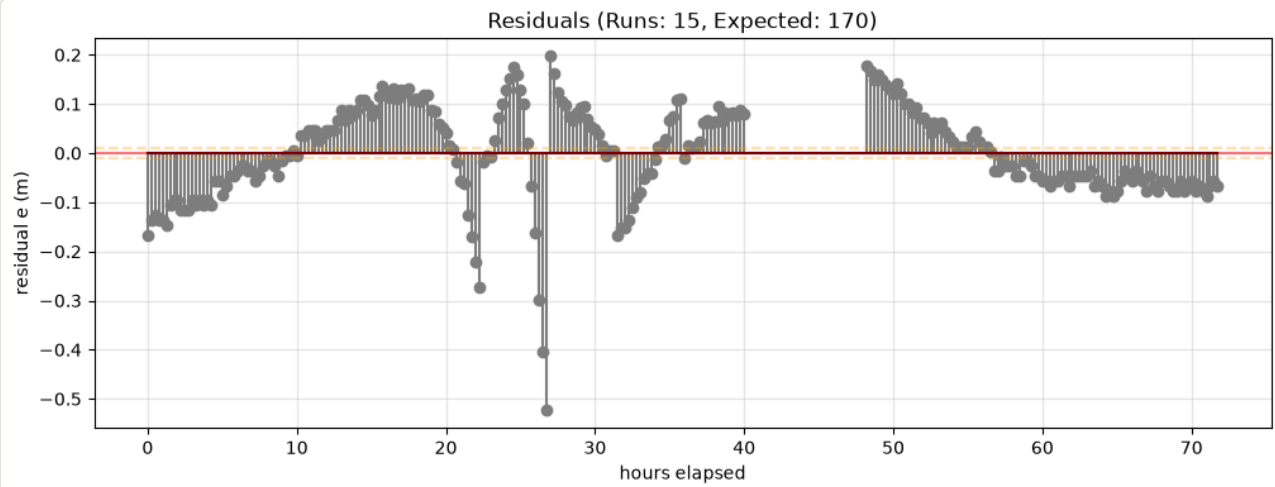
coefficient of determination

S

0.0972

m, standard error

Residuals



IN PLAIN WORDS

Residuals are the misses: actual reading minus predicted value. Positive means the curve sat too low there, negative means too high.

Sign runs: The residuals switch sign only **15** times. Under random noise, we'd expect about **170** switches. This pattern means the model is missing some real structure in the data.

Verdict. $R^2 = 0.998660$ and $s = 0.0972$ m. Trust this fit for volumes and timing to about a decimetre. Do not trust it for the level at an instant to the centimetre.